Superpopulation SE report

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I ran a simulation where I generated a fixed set of districts, and then repeatedly sampled clusters within those districts and conducted a cluster randomized experiment. I then looked at how much the point estimates varied across the simulation runs for all of our methods.

I nested the simulations in that I first generated a dataset (using the fixed set of district characteristics), and then repeatedly randomized clusters and estimated impacts. This gives us finite-sample standard errors for the estimators, keeping the full set of districts, clusters, and individuals fixed. We can also look at the standard deviation of the point estimates across all the simulations to get the superpopulation standard errors. This is a model where the clusters are sampled within districts, and individuals are sampled within clusters, but the districts are fixed.

The set of districts

To run a simulation scenario I generate a single set of districts. I use the "empirical" setting to make sure our multivariate normal distributions for the amount of cross-district impact variation and so forth equals our set values. I am therefore not worried about the simulation results being hugely dependent on *which* districts we have generated, as I expect all sets generated, given a specific set of parameters, would lead to the same final results.

The following is a set of descriptive features of the fixed set of districts in our simulation:

```
## K=20 J.bar=9.45 (3.22)   n.bar=20.44 (14.05) p.bar=0.53 (0.20)
## District stats:
##  district ATE=0.20 (0.10)    district ICC=0.44
##  cor(J,tau)=-0.75        cor(ptx,tau)=0.85
##  cor(J,alpha)=-0.81      cor(nbar, J)=-0.83
## Cluster stats:
##  school ICC=0.20 (0.00)
##  avg cluster size = 20.8 (range 1--141, sd = 23.1)
##  cor(n,tau) = 0.48 (across blocks)
## ATE information:
##  Within-district cluster ATE and variation: 0.20 (0.44)
##  Total cluster ATE: 0.18 (0.46)
##  Total individual ATE: 0.42 (0.44)
##
##   * Some stats estimated on 78454 generated observations across 3780 clusters.
```

The district ATE is the simple average of the district-level cluster-average ATEs. The bottom line are ATEs for a large dataset generated for this set of districts. We have good separation between cluster-average and individual-average ATE. We also see fairly high correlations of all sorts of features of the data, and all cluster sizes, etc., are varying.

The numbers in parentheses are the standard deviations of a given value. For example, J.bar, the average number of clusters per district, is around 9 and we have some districts with more and some with fewer (sd of

around 3). Note the district ATE and district impact variation are 0.20 and 0.10–exactly as we specified in the DGP call; this is the empirical setting doing its work.

Simulation results

For each estimator, we calculate the average estimated standard error, the average estimate, the bias, the RMSE, and the coverage. We calculate the bias, RMSE and coverage with respect to the appropriate estimand (Cluster or Person weighted).

The following table and figure shows our results. The table shows a variety of statistics, including the average of the degrees of freedom that we calculate for each estimator. In this simulation we are well away from the problematic behavior we have been seeing with the interacted estimators. The figure shows average estimate, the bias (relative to cluster or person average ATE as appropriate), the true standard error, and the RMSE. The whiskers are monte carlo standard errors, showing our uncertainty given the 500 iterations run.

Method	Weight	$\overline{\widehat{SE}}$	$\overline{\widehat{ATE}}$	Bias	RMSE	Coverage	% undef	df	SE_f	SE_s
MLM-RE	Cluster	0.10	0.25	0.08	0.12	0.88	0.00	167	0.09	0.08
MLM-FE	Cluster	0.10	0.22	0.05	0.10	0.94	0.00	159	0.09	0.08
MLM-FIcw	Cluster	0.10	0.20	0.03	0.10	0.95	0.00	149	0.10	0.09
LRa-FIcw-db	Cluster	0.10	0.17	-0.00	0.10	0.96	0.00	149	0.10	0.09
LRa-FIcw-het	Cluster	0.10	0.17	-0.00	0.10	0.96	0.00	149	0.10	0.09
LRicw-FIcw-db	Cluster	0.10	0.17	-0.00	0.10	0.96	0.00	149	0.10	0.09
LRa-FE-db	Cluster	0.10	0.18	0.00	0.09	0.97	0.00	168	0.10	0.09
LRa-FE-het	Cluster	0.10	0.18	0.00	0.09	0.97	0.00	168	0.10	0.09
LRicw-FE-db	Cluster	0.10	0.18	0.00	0.09	0.97	0.00	168	0.10	0.09
LRapw-FIpw-db	Person	0.09	0.39	-0.03	0.12	0.88	0.00	149	0.12	0.11
LRi-FIpw-db	Person	0.09	0.39	-0.03	0.12	0.88	0.00	149	0.12	0.11
LRapw-FIpw-het	Person	0.10	0.39	-0.03	0.12	0.92	0.00	149	0.12	0.11
LRi-FIpw-crve	Person	0.10	0.39	-0.03	0.12	0.92	0.00	37	0.12	0.11
LRapw-FE-db	Person	0.11	0.38	-0.03	0.12	0.93	0.00	168	0.12	0.11
LRi-FE-db	Person	0.11	0.38	-0.03	0.12	0.93	0.00	168	0.12	0.11
LRapw-FE-het	Person	0.12	0.38	-0.03	0.12	0.94	0.00	168	0.12	0.11
LRi-FE-crve	Person	0.12	0.38	-0.03	0.12	0.94	0.00	52	0.12	0.11

Table 1: Simulation 1 results



We are seeing general consistency across all the methods for the standard errors. For this DGP, the individual-average seems harder to estimate than the cluster average, likely due to the massive variation in cluster size coupled with the cluster variation, making the weighted average of clusters more unstable than the simple average.

From the table, we also see coverage can be low for some of the individual average estimators. Given that the degrees of freedom is fairly high, this is a disturbing result.

The low coverage may partly be driven by the small bias term of around -0.03 for the person-weighted estimators. The monte carlo standard errors show that this bias is real, and not just a fluke of the simulation (see the confidence intervals for the amount of bias in the figure). We also see real bias for the MLM estimators, which is their shrinking towards the person average effect.

Calibration of Estimated Standard Errors

We can look at how the average estimated standard error compares to both the true superpopulation standard error as well as the average of the true finite population standard error across the different datasets (first and second column). We can also look at how much larger superpopulation standard errors are from finite population (third column).

method	weight	SErat.s	SErat.f	SE_fs	bias	cover
LRi-FIpw-db	Person	0.77	0.85	1.11	-0.03	0.88
LRapw-FIpw-db	Person	0.77	0.85	1.11	-0.03	0.88
LRapw-FIpw-het	Person	0.85	0.94	1.11	-0.03	0.92
LRi-FIpw-crve	Person	0.85	0.94	1.11	-0.03	0.92
LRi-FE-db	Person	0.89	0.97	1.09	-0.03	0.93
LRapw-FE-db	Person	0.89	0.97	1.09	-0.03	0.93
LRapw-FE-het	Person	0.97	1.06	1.09	-0.03	0.94
LRi-FE-crve	Person	0.97	1.06	1.09	-0.03	0.94
MLM-FIcw	Cluster	1.00	1.11	1.10	0.03	0.95
LRicw-FIcw-db	Cluster	1.01	1.11	1.10	0.00	0.96
LRa-FIcw-db	Cluster	1.01	1.11	1.10	0.00	0.96

method	weight	SErat.s	SErat.f	SE_fs	bias	cover
LRa-FIcw-het	Cluster	1.01	1.11	1.10	0.00	0.96
LRa-FE-db	Cluster	1.02	1.14	1.11	0.00	0.97
LRicw-FE-db	Cluster	1.02	1.14	1.11	0.00	0.97
LRa-FE-het	Cluster	1.02	1.14	1.11	0.00	0.97
MLM-RE	Cluster	1.02	1.15	1.13	0.08	0.88
MLM-FE	Cluster	1.02	1.15	1.12	0.05	0.94

First, despite lots of heterogeneity at the cluster and district level, the finite and superpopulation true standard errors are all about the same. In simple blocked randomized (multisite) experiments, the relative inflation of the finite population standard error estimator (vz. true finite sample variance) is proportional to the treatment variation within the blocks. Cluster randomized experiments appear to have a different story, and it appears as if major differences in finite vs superpopulation standard error are unlikely to occur; see the other memo on this for more details.

Second, we are seeing how several individual-weight linear regression methods are systematically giving standard errors *much* too small in our context. The interacted estimators appear to be particularly vulnerable to this. We also see that coverage is fairly reasonable if the SE is fairly reasonable (e.g., LRi-FE-crve or LRapw_FE-het), despite the bias term. In other words, the bias does not seem to be driving the low coverage rates.

A second simulation (no correlation)

Repeating the above displays, we can look at the same simulation run on a dataset with the same variation in block and cluster sizes, but no strong correlation of size with impact or anything else.

The data were generated with complete independence, but we still see some non-zero correlations due to the random chance of generating the 20 districts (for reference, the correlation of 20 observations of two independent random normal variables has a 50% chance of being larger than 0.15 in magnitude, so correlations of this size are not unusual).

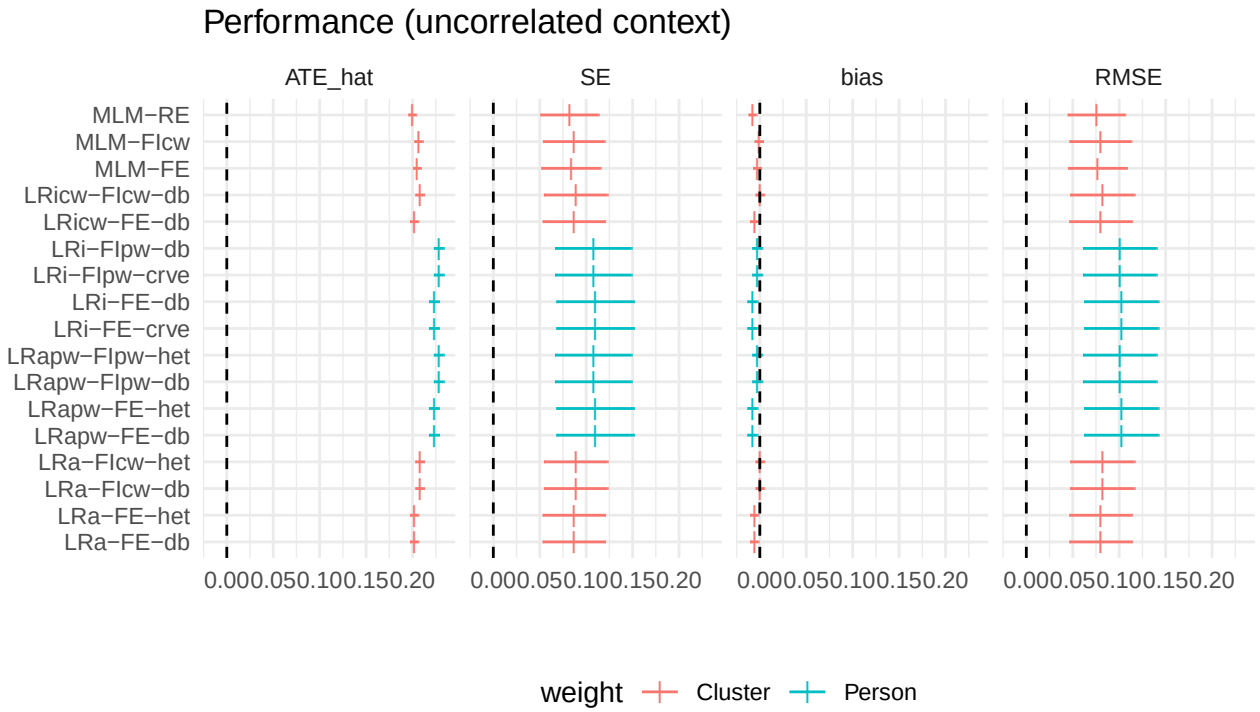
Descriptive statistics:

```
## K=20 J.bar=11.75 (4.40)  n.bar=33.77 (18.83) p.bar=0.50 (0.19)
## District stats:
##  district ATE=0.20 (0.10)    district ICC=0.45
##  cor(J,tau)=0.21     cor(ptx,tau)=-0.13
##  cor(J,alpha)=0.02       cor(nbar, J)=0.00
## Cluster stats:
##  school ICC=0.20 (0.00)
##  avg cluster size = 32.9 (range 1--189, sd = 34.6)
##  cor(n,tau) = 0.06 (across blocks)
## ATE information:
##  Within-district cluster ATE and variation: 0.21 (0.44)
##  Total cluster ATE: 0.21 (0.45)
##  Total individual ATE: 0.24 (0.46)
##
##   * Some stats estimated on 100427 generated observations across 3055 clusters.
```

Simulation results

Method	Weight	$\overline{\widehat{SE}}$	$\overline{\widehat{ATE}}$	Bias	RMSE	Coverage	% undef	df	SE_f	SE_s
LRa-FIcw-db	Cluster	0.09	0.21	0.00	0.08	0.96	0.00	195	0.09	0.08
LRa-FIcw-het	Cluster	0.09	0.21	0.00	0.08	0.96	0.00	195	0.09	0.08
LRicw-FIcw-db	Cluster	0.09	0.21	0.00	0.08	0.96	0.00	195	0.09	0.08
MLM-FIcw	Cluster	0.09	0.21	-0.00	0.08	0.97	0.00	195	0.09	0.08
LRa-FE-db	Cluster	0.09	0.20	-0.01	0.08	0.97	0.00	214	0.09	0.08
LRa-FE-het	Cluster	0.09	0.20	-0.01	0.08	0.97	0.00	214	0.09	0.08
LRicw-FE-db	Cluster	0.09	0.20	-0.01	0.08	0.97	0.00	214	0.09	0.08
MLM-FE	Cluster	0.08	0.20	-0.00	0.08	0.97	0.00	208	0.08	0.08
MLM-RE	Cluster	0.08	0.20	-0.01	0.08	0.97	0.00	216	0.08	0.07
LRapw-FIpw-db	Person	0.09	0.23	-0.00	0.10	0.93	0.00	195	0.11	0.10
LRi-FIpw-db	Person	0.09	0.23	-0.00	0.10	0.93	0.00	195	0.11	0.10
LRapw-FIpw-het	Person	0.10	0.23	-0.00	0.10	0.95	0.00	195	0.11	0.10
LRi-FIpw-crve	Person	0.10	0.23	-0.00	0.10	0.95	0.00	45	0.11	0.10
LRapw-FE-db	Person	0.11	0.22	-0.01	0.10	0.95	0.00	214	0.11	0.10
LRi-FE-db	Person	0.11	0.22	-0.01	0.10	0.95	0.00	214	0.11	0.10
LRapw-FE-het	Person	0.11	0.22	-0.01	0.10	0.96	0.00	214	0.11	0.10
LRi-FE-crve	Person	0.11	0.22	-0.01	0.10	0.96	0.00	68	0.11	0.10

Table 3: Simulation 2 (no correlation) results



With the no correlation case, we are not seeing bias for any of the estimators.

Calibrated Standard Errors

method	weight	SErat.s	SErat.f	SE_fs	cover
LRapw-FIpw-db	Person	0.87	0.94	1.07	0.93
LRi-FIpw-db	Person	0.87	0.94	1.07	0.93
LRapw-FIpw-het	Person	0.94	1.01	1.07	0.95
LRi-FIpw-crve	Person	0.94	1.01	1.07	0.95
LRapw-FE-db	Person	0.96	1.04	1.09	0.95
LRi-FE-db	Person	0.96	1.04	1.09	0.95
LRa-FIcw-het	Cluster	0.98	1.06	1.08	0.96
LRicw-FIcw-db	Cluster	0.98	1.06	1.08	0.96
LRa-FIcw-db	Cluster	0.98	1.06	1.08	0.96
LRa-FE-db	Cluster	0.99	1.08	1.09	0.97
LRicw-FE-db	Cluster	0.99	1.08	1.09	0.97
MLM-FIcw	Cluster	0.99	1.08	1.08	0.97
LRa-FE-het	Cluster	0.99	1.08	1.09	0.97
LRapw-FE-het	Person	1.00	1.09	1.09	0.96
LRi-FE-crve	Person	1.00	1.09	1.09	0.96
MLM-FE	Cluster	1.01	1.10	1.10	0.97
MLM-RE	Cluster	1.01	1.12	1.10	0.97

Without correlation, standard errors can be a bit too small, but they are better than before. This is a puzzling result.

Simulation 3: a smaller size experiment

We then look at a smaller experiment, of around 30 clusters across 4 districts, to see how things change. We make sure no district has fewer than 4 clusters, which ensures all estimators have defined estimated standard errors.

Descriptive statistics:

```
## K=4  J.bar=8.00 (3.37)   n.bar=116.12 (61.75)    p.bar=0.41 (0.13)
## District stats:
##  district ATE=0.20 (0.20)    district ICC=0.45
##  cor(J,tau)=0.94     cor(ptx,tau)=-0.98
##  cor(J,alpha)=-0.87      cor(nbar, J)=0.85
## Cluster stats:
##  school ICC=0.20 (0.00)
##  avg cluster size = 111.3 (range 1--476, sd = 113.1)
##  cor(n,tau) = -0.30 (across blocks)
## ATE information:
##  Within-district cluster ATE and variation: 0.18 (0.45)
##  Total cluster ATE: 0.23 (0.48)
##  Total individual ATE: 0.09 (0.44)
##
##   * Some stats estimated on 96201 generated observations across 864 clusters.
```

Simulation results

Method	Weight	$\overline{\widehat{SE}}$	$\overline{\widehat{ATE}}$	Bias	RMSE	Coverage	% undef	df	SE_f	SE_s
MLM-FIcw	Cluster	0.20	0.24	-0.02	0.21	0.93	0.00	24	0.22	0.20
LRa-FIcw-db	Cluster	0.22	0.25	-0.00	0.21	0.94	0.00	24	0.22	0.21
LRa-FIcw-het	Cluster	0.22	0.25	-0.00	0.21	0.94	0.00	24	0.22	0.21
LRicw-FIcw-db	Cluster	0.22	0.25	-0.00	0.21	0.94	0.00	24	0.22	0.21
MLM-FE	Cluster	0.20	0.22	-0.04	0.21	0.94	0.00	27	0.21	0.20
MLM-RE	Cluster	0.20	0.25	-0.01	0.20	0.94	0.00	28	0.22	0.20
LRa-FE-db	Cluster	0.22	0.23	-0.03	0.21	0.95	0.00	27	0.22	0.20
LRicw-FE-db	Cluster	0.22	0.23	-0.03	0.21	0.95	0.00	27	0.22	0.20
LRa-FE-het	Cluster	0.22	0.23	-0.03	0.21	0.95	0.00	27	0.22	0.20
LRapw-FIpw-db	Person	0.22	0.17	0.05	0.30	0.81	0.00	24	0.31	0.29
LRi-FIpw-db	Person	0.22	0.17	0.05	0.30	0.81	0.00	24	0.31	0.29
LRapw-FIpw-het	Person	0.25	0.17	0.05	0.30	0.85	0.00	24	0.31	0.29
LRi-FIpw-crve	Person	0.25	0.17	0.05	0.30	0.85	0.00	6	0.31	0.29
LRapw-FE-db	Person	0.23	0.13	0.01	0.27	0.86	0.00	27	0.29	0.27
LRi-FE-db	Person	0.23	0.13	0.01	0.27	0.86	0.00	27	0.29	0.27
LRapw-FE-het	Person	0.26	0.13	0.01	0.27	0.90	0.00	27	0.29	0.27
LRi-FE-crve	Person	0.26	0.13	0.01	0.27	0.90	0.00	8	0.29	0.27

Table 5: Simulation 3 (small number blocks) results



Calibrated Standard Errors

method	weight	SErat.s	SErat.f	SE_fs	cover
LRapw-FIpw-db	Person	0.87	0.94	1.07	0.93
LRi-FIpw-db	Person	0.87	0.94	1.07	0.93
LRapw-FIpw-het	Person	0.94	1.01	1.07	0.95
LRi-FIpw-crve	Person	0.94	1.01	1.07	0.95
LRapw-FE-db	Person	0.96	1.04	1.09	0.95
LRi-FE-db	Person	0.96	1.04	1.09	0.95
LRa-FIcw-het	Cluster	0.98	1.06	1.08	0.96
LRicw-FIcw-db	Cluster	0.98	1.06	1.08	0.96
LRa-FIcw-db	Cluster	0.98	1.06	1.08	0.96
LRa-FE-db	Cluster	0.99	1.08	1.09	0.97
LRicw-FE-db	Cluster	0.99	1.08	1.09	0.97
MLM-FIcw	Cluster	0.99	1.08	1.08	0.97
LRa-FE-het	Cluster	0.99	1.08	1.09	0.97
LRapw-FE-het	Person	1.00	1.09	1.09	0.96
LRi-FE-crve	Person	1.00	1.09	1.09	0.96
MLM-FE	Cluster	1.01	1.10	1.10	0.97
MLM-RE	Cluster	1.01	1.12	1.10	0.97

The coverage of all the estimators is low, because the standard errors are generally too small.

The degrees of freedom seem to not correctly correspond to the different estimators. E.g., LRi-FE-CRVE has smaller df than the interacted estimators. I did some investigating and the low df for the LRi-FE-CRVE is indeed coming from the `lm_robust()` method, and is probably something to do with how degrees of freedom are calculated when the cluster sizes vary a lot. The ad hoc df calculations for the design-based estimators and interacted estimators, by contrast, just use total clusters minues number of estimated coefficients, and thus are stable to different cluster sizes. I am not sure if one df calculation vs. another is preferred.

Action Items

- POSSIBLE TODO: Ask Peter Schochet to weigh in on the df calculations for the db estimators after pointing out the df that comes out of the robust linear model package.

- POSSIBLE TODO: Try and track down how df are calculated by lm_robust with CRVE–and I expect the method of CRVE dictates df.
- POSSIBLE TODO: Ask James P. about df for robust standard errors. He seems to know this stuff.

Simulation 4 & 5: Exploring finite vs. superpopulation standard errors

Simulations 4 and 5 are designed to push the envelope on separating the finite and superpopulation standard errors, given the fixed districts. These two scenarios have the exact same district characteristics as simulation 3, except for changing the amount of within-district treatment variation to 0 or to a large amount.

Here are the stats:

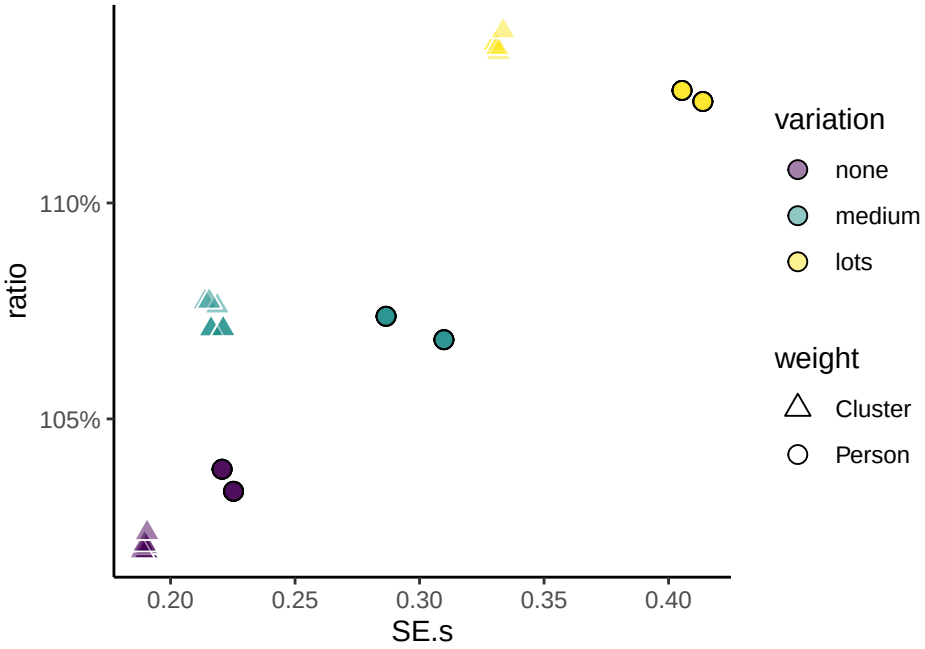
```
## K=4  J.bar=7.00 (1.15)   n.bar=48.64 (9.42)  p.bar=0.44 (0.15)
## District stats:
##  district ATE=0.20 (0.20)    district ICC=0.45
##  cor(J,tau)=-0.38        cor(ptx,tau)=-0.57
##  cor(J,alpha)=0.38       cor(nbar, J)=0.55
## Cluster stats:
##  school ICC=0.20 (0.00)
##  avg cluster size = 47.5 (range 1--171, sd = 37.7)
##  cor(n,tau) = -0.11 (across blocks)
## ATE information:
##  Within-district cluster ATE and variation: 0.20 (0.00)
##  Total cluster ATE: 0.19 (0.17)
##  Total individual ATE: 0.18 (0.16)
##
##   * Some stats estimated on 97190 generated observations across 2044 clusters.
```

```
## K=4  J.bar=7.00 (1.15)   n.bar=48.64 (9.42)  p.bar=0.44 (0.15)
## District stats:
##  district ATE=0.20 (0.20)    district ICC=0.45
##  cor(J,tau)=-0.38        cor(ptx,tau)=-0.57
##  cor(J,alpha)=0.38       cor(nbar, J)=0.55
## Cluster stats:
##  school ICC=0.20 (0.00)
##  avg cluster size = 49.5 (range 1--172, sd = 38.8)
##  cor(n,tau) = -0.04 (across blocks)
## ATE information:
##  Within-district cluster ATE and variation: 0.22 (0.90)
##  Total cluster ATE: 0.20 (0.91)
##  Total individual ATE: 0.18 (0.89)
##
##   * Some stats estimated on 101243 generated observations across 2044 clusters.
```

And now for our three cases, we can see that superpopulation SEs do get relatively larger than finite population as treatment variation increases, but the percent increase is not as large as one might expect.



Here are the ratios of super to finite true standard error vs. the true superpopulation standard error.



What is confusing is these experiments are modest in size, with around 30 clusters across 5 districts. So given our multi-site work, I would have thought to see greater differences in finite vs. superpopulation SE. The

separate memo on finite vs. superpopulation standard errors does provide some reassurance that these results are indeed correct, however.

Simulation 6: A sanity check simulation

In this simulation the cluster sizes do not vary much and basically we are in our (near) constant size everything. The question is whether the SEs are well behaved in this fairly nice case. We only have 30 clusters in this nice case, to try and keep within mostly realistic bounds of size.

Descriptive statistics:

```
## K=5  J.bar=6.00 (0.00)   n.bar=39.00 (0.00)  p.bar=0.50 (0.00)
## District stats:
##  district ATE=0.20 (0.10)    district ICC=0.00
##  cor(J,tau)=NA       cor(ptx,tau)=NA
##  cor(J,alpha)=NA     cor(nbar, J)=NA
## Cluster stats:
##  school ICC=0.20 (0.00)
##  avg cluster size = 39.0 (range 28--50, sd = 6.6)
##  cor(n,tau) = -0.01 (across blocks)
## ATE information:
##  Within-district cluster ATE and variation: 0.21 (0.20)
##  Total cluster ATE: 0.21 (0.22)
##  Total individual ATE: 0.21 (0.22)
##
##   * Some stats estimated on 93692 generated observations across 2400 clusters.
```

Simulation results

Method	Weight	$\overline{\widehat{SE}}$	$\overline{\widehat{ATE}}$	Bias	RMSE	Coverage	% undef	df	SE_f	SE_s
LRa-FIcw-db	Cluster	0.17	0.21	0.01	0.18	0.94	0.00	20	0.18	0.18
LRa-FIcw-het	Cluster	0.17	0.21	0.01	0.18	0.94	0.00	20	0.18	0.18
LRicw-FIcw-db	Cluster	0.17	0.21	0.01	0.18	0.94	0.00	20	0.18	0.18
MLM-RE	Cluster	0.17	0.21	0.01	0.18	0.94	0.00	26	0.18	0.18
MLM-FIcw	Cluster	0.17	0.21	0.01	0.18	0.94	0.00	20	0.18	0.18
LRa-FE-db	Cluster	0.18	0.21	0.01	0.18	0.94	0.00	24	0.18	0.18
LRa-FE-het	Cluster	0.18	0.21	0.01	0.18	0.94	0.00	24	0.18	0.18
LRicw-FE-db	Cluster	0.18	0.21	0.01	0.18	0.94	0.00	24	0.18	0.18
MLM-FE	Cluster	0.18	0.21	0.01	0.18	0.95	0.00	24	0.18	0.18
LRapw-FIpw-db	Person	0.17	0.21	0.01	0.18	0.93	0.00	20	0.18	0.18
LRi-FIpw-db	Person	0.17	0.21	0.01	0.18	0.93	0.00	20	0.18	0.18
LRapw-FIpw-het	Person	0.18	0.21	0.01	0.18	0.94	0.00	20	0.18	0.18
LRi-FIpw-crve	Person	0.18	0.21	0.01	0.18	0.94	0.00	19	0.18	0.18
LRapw-FE-db	Person	0.18	0.21	0.01	0.18	0.94	0.00	24	0.18	0.18
LRi-FE-db	Person	0.18	0.21	0.01	0.18	0.94	0.00	24	0.18	0.18
LRapw-FE-het	Person	0.18	0.21	0.01	0.18	0.94	0.00	24	0.18	0.18
LRi-FE-crve	Person	0.18	0.21	0.01	0.18	0.94	0.00	23	0.18	0.18

Table 7: Simulation 2 (no correlation) results



Calibrated Standard Errors

method	weight	SErat.s	SErat.f	SE_fs	cover
LRi-FIpw-db	Person	0.95	0.96	1.01	0.93
LRapw-FIpw-db	Person	0.95	0.96	1.01	0.93
MLM-RE	Cluster	0.96	0.97	1.01	0.94
LRapw-FIpw-het	Person	0.96	0.97	1.01	0.94
LRi-FIpw-crve	Person	0.96	0.97	1.01	0.94
LRicw-FIcw-db	Cluster	0.97	0.98	1.01	0.94
LRa-FIcw-het	Cluster	0.97	0.98	1.01	0.94
LRa-FIcw-db	Cluster	0.97	0.98	1.01	0.94
MLM-FIcw	Cluster	0.97	0.98	1.01	0.94
LRapw-FE-db	Person	0.97	0.98	1.01	0.94
LRi-FE-db	Person	0.97	0.98	1.01	0.94
LRa-FE-het	Cluster	0.97	0.99	1.01	0.94
LRa-FE-db	Cluster	0.97	0.99	1.01	0.94
LRicw-FE-db	Cluster	0.97	0.99	1.01	0.94
MLM-FE	Cluster	0.97	0.99	1.01	0.95
LRi-FE-crve	Person	0.97	0.98	1.01	0.94
LRapw-FE-het	Person	0.97	0.98	1.01	0.94

Everything is estimating the same thing, estimated standard errors are mostly calibrated (but a bit small, on average), and coverage is good.

Simulation 8: A second sanity check simulation

Here we replicate simulation 3 (of 30 clusters), but just restrict cluster sizes, but we still have all the correlations.

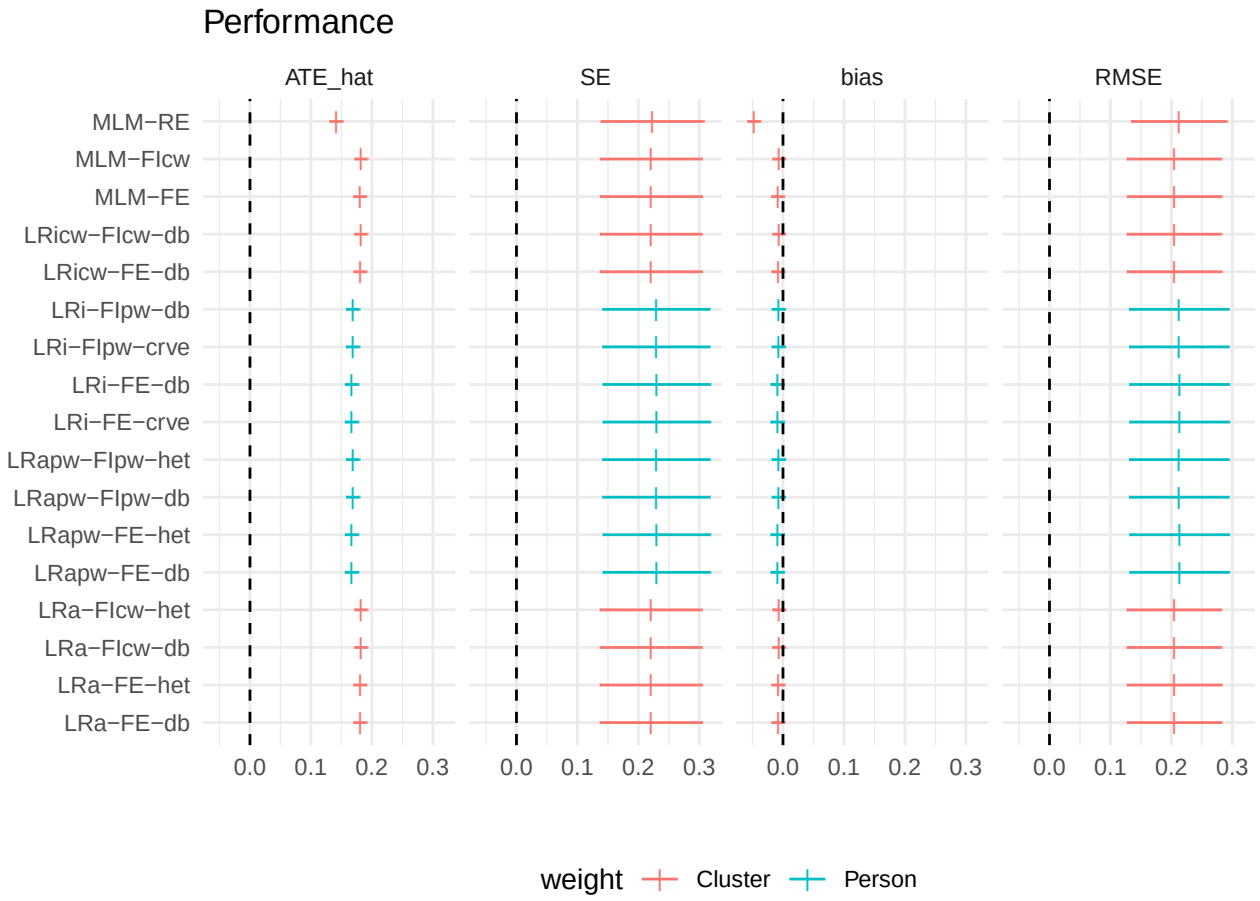
Descriptive statistics:

```
## K=4  J.bar=7.00 (1.15)   n.bar=48.64 (9.42)  p.bar=0.44 (0.15)
## District stats:
##  district ATE=0.20 (0.20)    district ICC=0.45
##  cor(J,tau)=-0.38        cor(ptx,tau)=-0.57
##  cor(J,alpha)=0.38       cor(nbar, J)=0.55
## Cluster stats:
##  school ICC=0.20 (0.00)
##  avg cluster size = 48.8 (range 30--80, sd = 12.1)
##  cor(n,tau) = -0.07 (across blocks)
## ATE information:
##  Within-district cluster ATE and variation: 0.20 (0.46)
##  Total cluster ATE: 0.19 (0.48)
##  Total individual ATE: 0.18 (0.48)
##
##   * Some stats estimated on 99743 generated observations across 2044 clusters.
```

Simulation results

Method	Weight	$\overline{\widehat{SE}}$	$\overline{\widehat{ATE}}$	Bias	RMSE	Coverage	% undef	df	SE_f	SE_s
MLM-RE	Cluster	0.21	0.14	-0.05	0.21	0.94	0.00	23	0.22	0.21
MLM-FIcw	Cluster	0.21	0.18	-0.01	0.20	0.94	0.00	20	0.22	0.21
MLM-FE	Cluster	0.21	0.18	-0.01	0.20	0.95	0.00	23	0.22	0.21
LRa-FIcw-db	Cluster	0.22	0.18	-0.01	0.20	0.95	0.00	20	0.22	0.21
LRa-FIcw-het	Cluster	0.22	0.18	-0.01	0.20	0.95	0.00	20	0.22	0.21
LRicw-FIcw-db	Cluster	0.22	0.18	-0.01	0.20	0.95	0.00	20	0.22	0.21
LRa-FE-db	Cluster	0.22	0.18	-0.01	0.20	0.95	0.00	23	0.22	0.21
LRicw-FE-db	Cluster	0.22	0.18	-0.01	0.20	0.95	0.00	23	0.22	0.21
LRa-FE-het	Cluster	0.22	0.18	-0.01	0.20	0.96	0.00	23	0.22	0.21
LRapw-FIpw-db	Person	0.22	0.17	-0.01	0.21	0.94	0.00	20	0.23	0.21
LRi-FIpw-db	Person	0.22	0.17	-0.01	0.21	0.94	0.00	20	0.23	0.21
LRapw-FIpw-het	Person	0.22	0.17	-0.01	0.21	0.94	0.00	20	0.23	0.21
LRi-FIpw-crve	Person	0.22	0.17	-0.01	0.21	0.94	0.00	12	0.23	0.21
LRapw-FE-db	Person	0.23	0.17	-0.01	0.21	0.95	0.00	23	0.23	0.21
LRi-FE-db	Person	0.23	0.17	-0.01	0.21	0.95	0.00	23	0.23	0.21
LRapw-FE-het	Person	0.23	0.17	-0.01	0.21	0.95	0.00	23	0.23	0.21
LRi-FE-crve	Person	0.23	0.17	-0.01	0.21	0.95	0.00	16	0.23	0.21

Table 9: Simulation 8 (small variablity) results



Calibrated Standard Errors

method	weight	SErat.s	SErat.f	SE_fs	cover
LRi-FIpw-db	Person	0.95	0.96	1.01	0.93
LRapw-FIpw-db	Person	0.95	0.96	1.01	0.93
MLM-RE	Cluster	0.96	0.97	1.01	0.94
LRapw-FIpw-het	Person	0.96	0.97	1.01	0.94
LRi-FIpw-crve	Person	0.96	0.97	1.01	0.94
LRicw-FIcw-db	Cluster	0.97	0.98	1.01	0.94
LRa-FIcw-het	Cluster	0.97	0.98	1.01	0.94
LRa-FIcw-db	Cluster	0.97	0.98	1.01	0.94
MLM-FIcw	Cluster	0.97	0.98	1.01	0.94
LRapw-FE-db	Person	0.97	0.98	1.01	0.94
LRi-FE-db	Person	0.97	0.98	1.01	0.94
LRa-FE-het	Cluster	0.97	0.99	1.01	0.94
LRa-FE-db	Cluster	0.97	0.99	1.01	0.94
LRicw-FE-db	Cluster	0.97	0.99	1.01	0.94
MLM-FE	Cluster	0.97	0.99	1.01	0.95
LRi-FE-crve	Person	0.97	0.98	1.01	0.94
LRapw-FE-het	Person	0.97	0.98	1.01	0.94

Everything is estimating the same thing, estimated standard errors are mostly calibrated (but a bit small, on average), and coverage is good.

Comparing Simulation 3 to Simulation 8

Simulation 3 and 8 differ *only* in terms of the variation in cluster size. This has various downstream effects as well.

method	bias_3	bias_8	SE.s_3	SE.s_8	RMSE_3	RMSE_8	cover_3	cover_8
LRa-FE-db	-0.03	-0.01	0.22	0.22	0.21	0.20	0.95	0.95
LRa-FE-het	-0.03	-0.01	0.22	0.22	0.21	0.20	0.95	0.96
LRa-FIcw-db	0.00	-0.01	0.22	0.22	0.21	0.20	0.94	0.95
LRa-FIcw-het	0.00	-0.01	0.22	0.22	0.21	0.20	0.94	0.95
LRapw-FE-db	0.01	-0.01	0.29	0.23	0.27	0.21	0.86	0.95
LRapw-FE-het	0.01	-0.01	0.29	0.23	0.27	0.21	0.90	0.95
LRapw-FIpw-db	0.05	-0.01	0.31	0.23	0.30	0.21	0.81	0.94
LRapw-FIpw-het	0.05	-0.01	0.31	0.23	0.30	0.21	0.85	0.94
LRi-FE-crve	0.01	-0.01	0.29	0.23	0.27	0.21	0.90	0.95
LRi-FE-db	0.01	-0.01	0.29	0.23	0.27	0.21	0.86	0.95
LRi-FIpw-crve	0.05	-0.01	0.31	0.23	0.30	0.21	0.85	0.94
LRi-FIpw-db	0.05	-0.01	0.31	0.23	0.30	0.21	0.81	0.94
LRicw-FE-db	-0.03	-0.01	0.22	0.22	0.21	0.20	0.95	0.95
LRicw-FIcw-db	0.00	-0.01	0.22	0.22	0.21	0.20	0.94	0.95
MLM-FE	-0.04	-0.01	0.21	0.22	0.21	0.20	0.94	0.95
MLM-FIcw	-0.02	-0.01	0.22	0.22	0.21	0.20	0.93	0.94
MLM-RE	-0.01	-0.05	0.22	0.22	0.20	0.21	0.94	0.94

